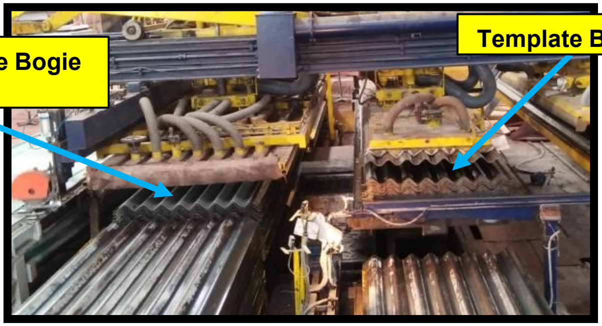
QUALITY CIRCLE- (8D Approach)

PROJECT NAME	:	Green sheet Recovery system during Template bogie changeover to avoid Green sheet wastage.
PLANT(Location)	:	Podanur Works
DATE OF START OF PROJECT	:	November 2017
DATE OF COMPLETION OF PROJECT	:	April 2018

Project Title & Team Formation

Project Title: Green sheet recovery system during template bogie changeover to avoid Green sheet wastage.

Implemented Area: Sheeting Machine -2 Shop floor, Podanur Works, Coimbatore



Cross Functional Team:

Mr. M. Vijay Kumar
Electrical Maint.

Mr. K. Senthil Kumar
Electrical Supervisor
Maint.

Mr. Srinivasagam
Electrician

Mr. A. N. Vignesh
Prod.

Team Leader
Mr. C. Murugesan
(Sr. Manager, Maint.)

Mr. Devaraj
Fitter

Mr. Suresh
Fitter

Mr. Murali
Fitter

Mr. Pillai
Fabricator

Mr. Shamugaraj
Board Operator

Project Description

Green sheets recovery system during Template bogie changeover to avoid Green sheets wastage.
(Target:- To eliminate Green sheets wastage)

Reason for selecting THEME :

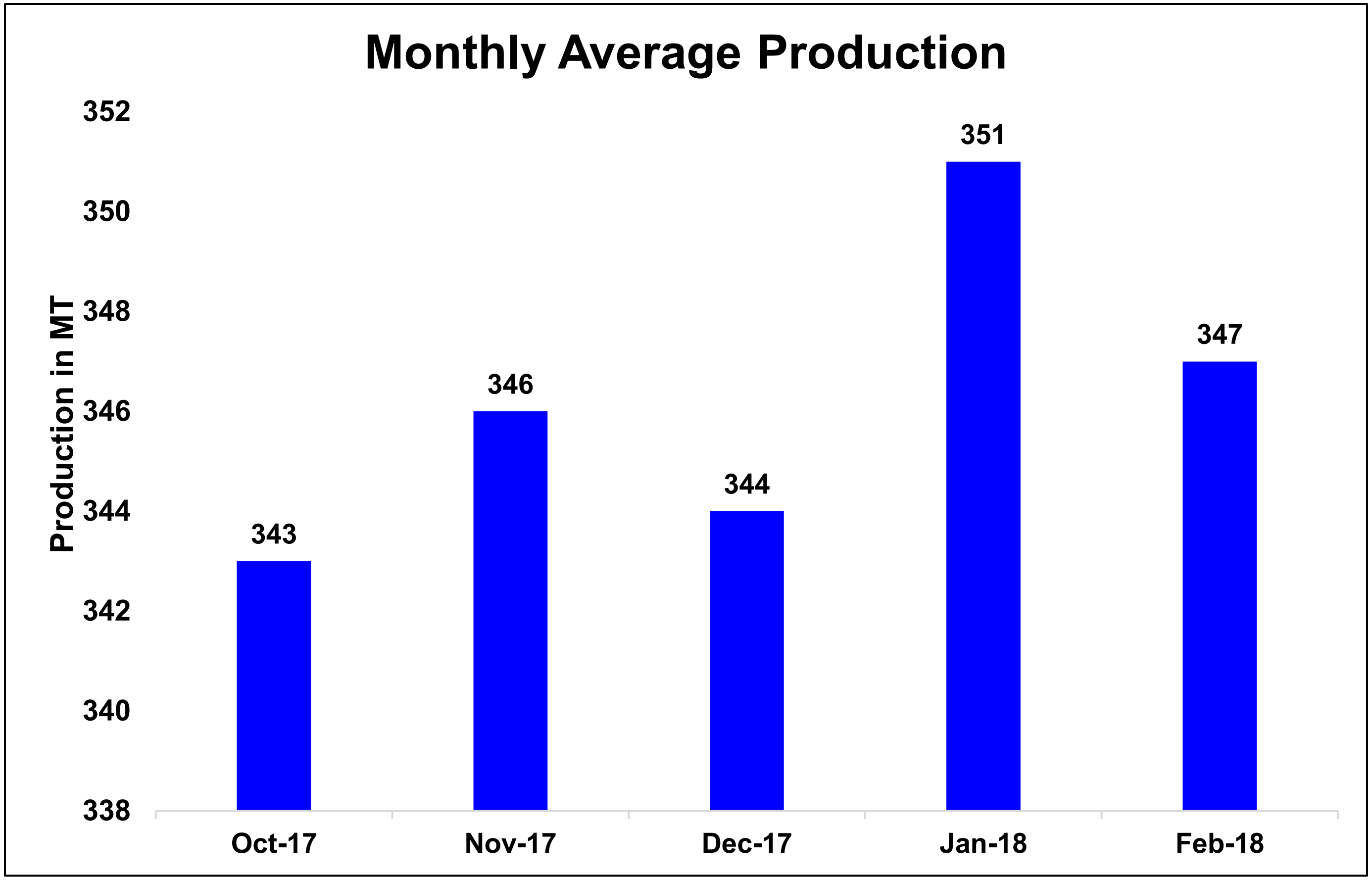
- ***Two Green sheets wastage per cycle during Template bogie changeover.***
- ***Production loss of 14 MT/Day.***
- ***Revenue loss of 1.4 Lakh/Day.***

Problem :

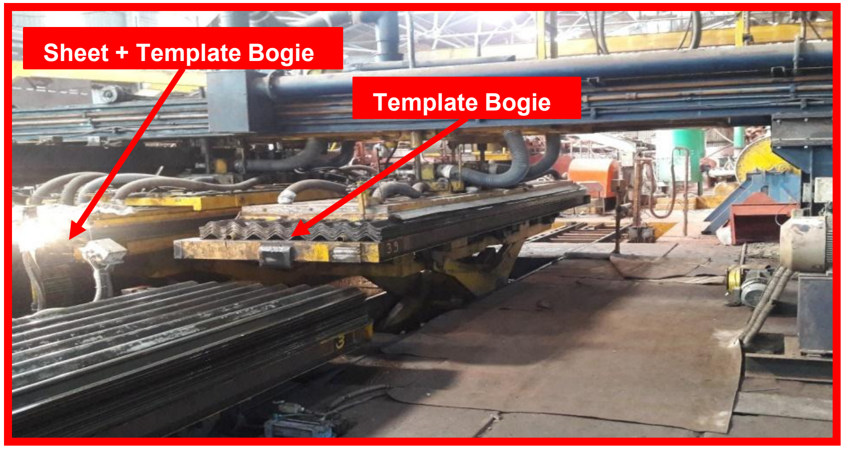
Due to Non-availability of Metal Templates for Piling mixed bogie during Template Bogie changeover there is Loss of Green sheets(2 sheets/cycle). This is a potential loss of approx. 3.5% which if avoided can leads to productivity improvement.

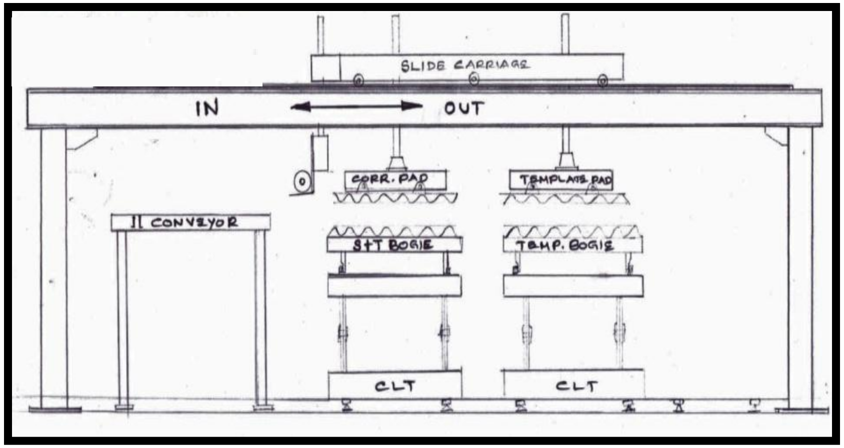
Existing Data – PW2 Monthly Production Trend

- Previous Monthly production trend.



Problem Description: Before situation





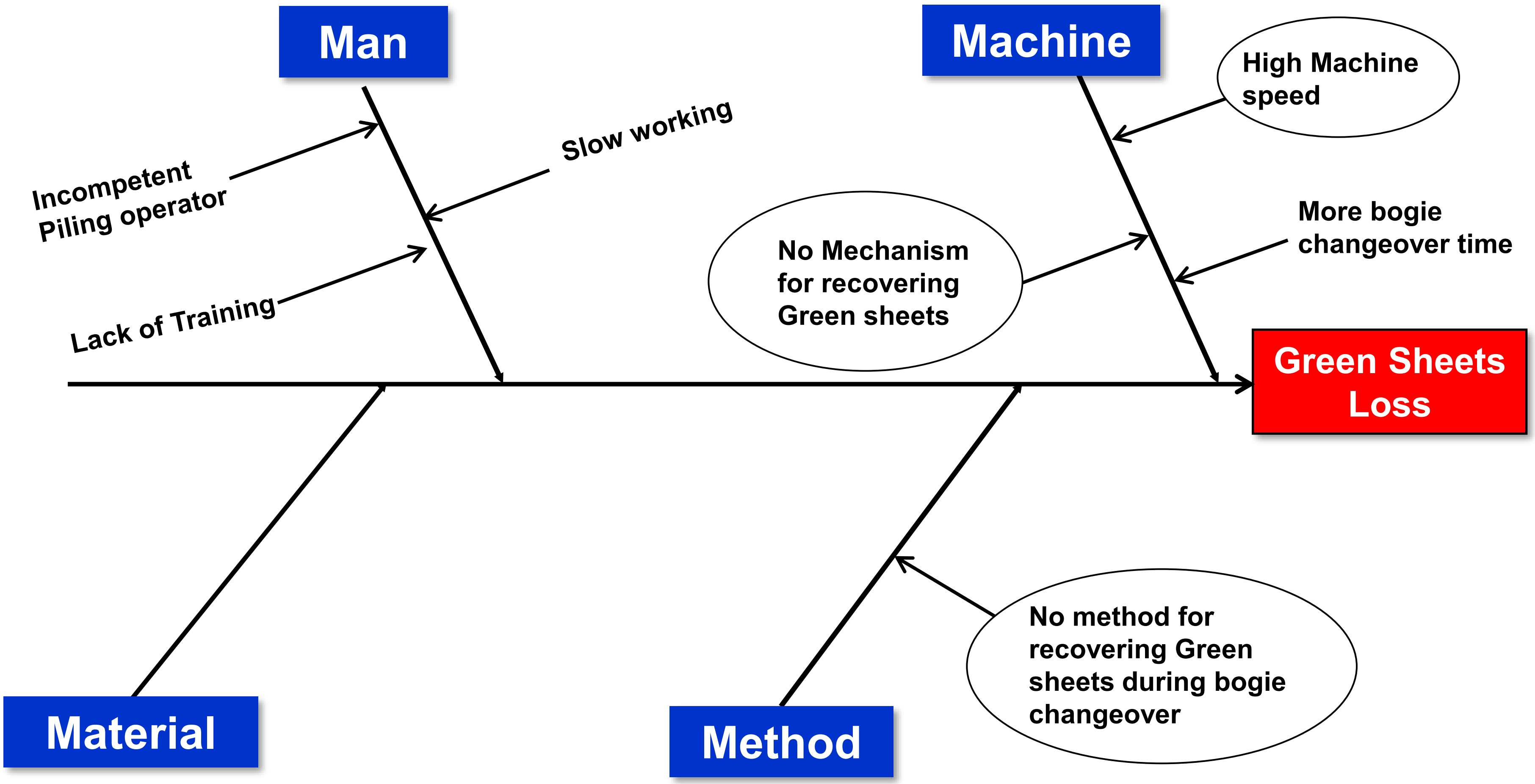
- There was no additional Template pad for recovering templates to stack shuttle trolley with templates.
- There was no side carriage to mount template pilling pad.
- There was no shuttle trolley for stacking recovered templates from template bogie.
- There was no Rail Line & Pneumatic cylinder arrangement for movement of Shuttle trolley IN & OUT.

STEP – 3 INTERIMN CONTAINMENT ACTIONS

EVEREST

S.NO	CONTAINMENT ACTIONS	RESPONSIBILITY	TARGET DATE
1	Slide carriage to be extended with wheel for mounting of Additional template pad with guided assembly.	Maintenance In-charge	15/01/18
2	Additional Template pad to be provided with Pneumatic cylinder for collecting templates from template bogie and transfer the same to shuttle trolley to meet the template requirement during bogie changeover.	Maintenance In-charge	31/01/18
3	New shuttle trolley to be fabricated and provided for template stock and utilizing it during Template Bogie changeover time.	Maintenance In-charge	15/02/18
4	Rail line with Pneumatic cylinder to be provided for shuttle trolley IN & OUT movement.	Maintenance In-charge	28/02/18
5	To provide PLC controls for the system.	Electrical Maintenance In-charge	09/03/2018

Cause & Effect Analysis



WHY WHY ANALYSIS

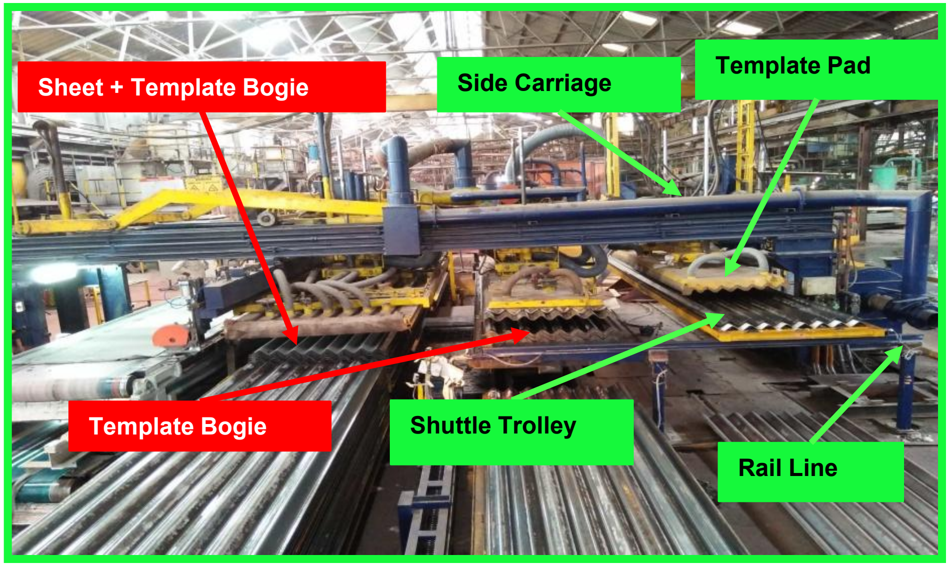
EVEREST

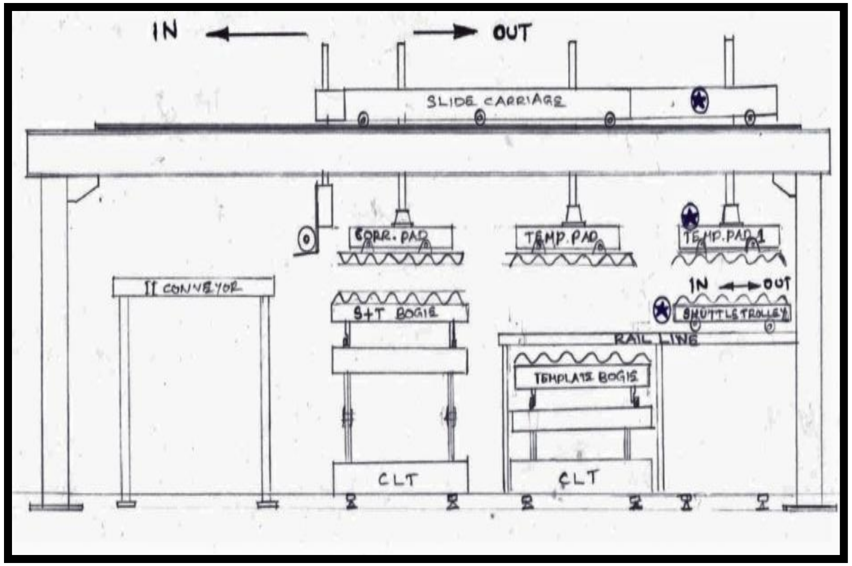
Problem Statement	Loss of Production
1st why?	Why there is Loss of Production?
Answer:	Loss of Green sheets during Template bogie changeover.
2nd why?	Why there is Loss of Green sheets during Template bogie changeover?
Answer:	During Template bogie changeover there no Templates to provide for mixed bogie piling. So that, we can't pile Green sheets in Mixed bogie.
3rd why	Why there are no Templates for Mixed bogie piling?
Answer:	For Template bogie changeover it takes some time and during that Templates will be not available.
Root Cause:	**Design constraint of the Piling unit.**
Solution	**To provide Additional, rolling Templates storage pad for storing Templates & Template piling pad for piling templates in Templates storage pad.**

Permanent Corrective Actions

S. No	Actions	Resp.	Target Date	Status
1	Slide carriage is extended with wheel for mounting of Additional template pad with guided assembly.	Maintenance In-charge	15/01/18	Completed
2	Additional Template pad is provided with Pneumatic cylinder for collecting templates from template bogie and transfer the same to shuttle trolley to meet the template requirement during bogie changeover.	Maintenance In-charge	31/01/18	Completed
3	New shuttle trolley is fabricated and provided for template stock and utilizing it during Template Bogie changeover time.	Maintenance In-charge	15/02/18	Completed
4	Rail line with Pneumatic cylinder is provided for shuttle trolley IN & OUT movement.	Maintenance In-charge	28/02/18	Completed
5	Provided PLC controls for operating the system.	Electrical Maintenance In-charge	10/03/2018	Completed

Permanent Corrective Actions





- Side carriage is made to mount template pilling pad.
- Template pilling pad is fabricated and installed to side carriage.
- Shuttle trolley is made and installed on Rail Line with Pneumatic cylinder arrangement for movement of Shuttle trolley IN & OUT.
- The entire system is automatized for continuous operation.
- Due to this system there is production saving of 14 MT/day.

Permanent Corrective Actions

- Implementation of Permanent corrective action as kaizen.

KAIZEN SHEET	Department	Maintenance	Works	BW	SW	CW	KW	LW	PW	NW	BW- ESBS	EVEREST
Kaizen Number	Line / Section	SM2	Result	P		Q		C	D	S	M	
PW/KZN/04/18	Machine /Equipment	Pilling		Productivity		Quality		Cost	Delivery	Safety	Morale	QS/KZN/01

KAIZEN DESCRIPTION : Green sheets recovery system during Template bogie changeover.

IDEA / Thought : To provide additional Template pad and Template pilling suction pad to provide templates to mixed bogie during Template bogie changeover.

Problem / Present Status	Counter Measure
Previously there is loss of Greensheets during Template Bogie changeover.	Additional rolling Template supply pad and Template pilling pad provided to provide templates for mixed bogie pilling during Template bogie changeover.

Target	
Kaizen Start Date	01-Oct-17
Kaizen Finish Date	09-Mar-18

BEFORE

AFTER

Team Leader

C. Murgesan(Sr. Manager Maintenance)

Team Members :

M. Vijay Kumar (Electrical Engineer), K. Senthil Kumar(Supervisor), Srinivasagam(Electrician), Devaraj(Fitter), Pillai(Fabricator), Murali(Fitter), Suresh(Fitter)

Investment cost : Rs. 80,000/-

Benefits :

Productivity improvement by saving Green sheet during template bogie changeover.

Daily no of boges pilled **per shift** = 31 & No.of sheets saving per bogie = **2**

For **1 Day = 31 x 3 x 2 x 6 x 12. 59 = 14.05 MT/Day**

For **25 Days in month** = **14.05 x 25** = **351 MT/Month**

For **1 year production saving** = Nearly **4200 MT/Year**

KAIZEN SUSTAINANCE :

WHAT TO DO	Regular inspection and maintenance
HOW TO DO	Giving taining and working instructions to Maintenance person.
FREQUENCY	Daily

Why Why Analysis

Problem :- During Template bogie changeover there is loss of green sheets.

Why 1. There are no templates to provide for mixed bogie pilling during template bogie changeover.

Why 2. It takes some time for template bogie changeover and at that time there is no mechanism for providing templates.

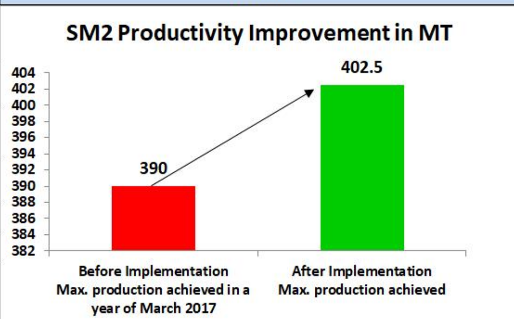
Why 3.

Why 4.

Why 5.

Action Taken : Additional template storage pad and template pilling pad provided to provide templates to mixed bogie during template bogie changeover.

GRAPHS

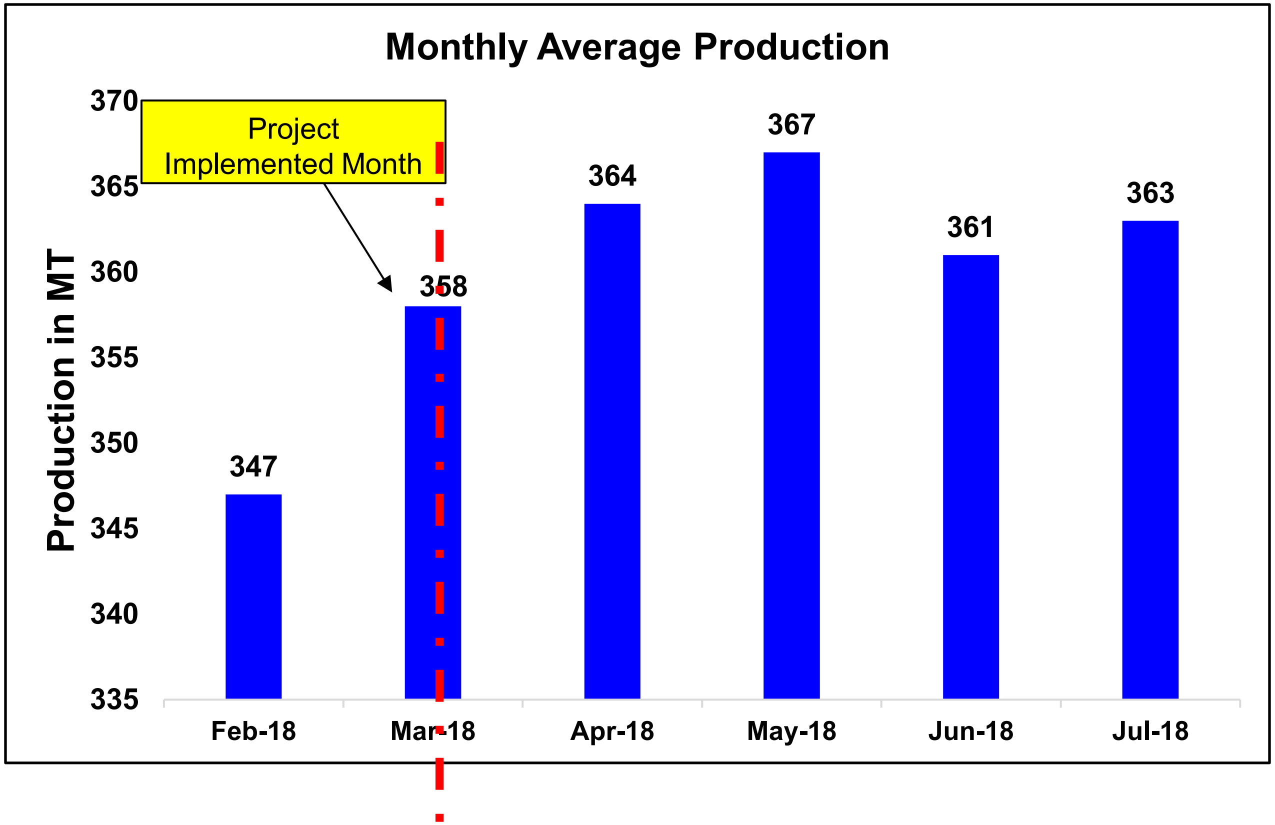


Scope Of Horizontal Deployment

BW-ESBS	NW	BW	CW	KW	LW	PW	SW
		√	√	√	√	√	√
Equipment		Target Date				Resp	Status

Validation of Data

- Monthly Average Production trend after implementing permanent corrective action.



Validation of Permanent Corrective Actions

EVEREST

Tangible Benefits

- Our Productivity increased by 14MT/Day.
- Our Max. Production increased from 390 MT/Day to 402.5 MT/Day.
- There is Production saving of approx. 4500 MT/Year.
- Revenue saving of 4.53 Cr./Annum.
- Potential cost saving of 1.38 Cr./Annum.
- Quality of our sheets improved due to reduction in recycled waste by 18%.

In-tangible Benefits

- Morale of people increased due to increase in Productivity.
- Corrective action which we done it as a kaizen won the best kaizen award in Q3 inter works competition.

Standardization

- Training on Standard Operating Procedures are given to all the Workmen & maintenance people to operate the system and to perform regular inspection & maintenance.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Congratulate the team

Yes / No	Signature / Title / Date	Findings
Yes	M. Gnana Prakash	Problem solving is an effective tool which is used for knowing the hidden facts to improve Productivity. Further this process can be used in other area for solving the problems.

Thank You